

KISAN MITRA • AI CROP DIAGNOSIS REPORT

Wheat Aphid Infestation — likely Green Aphid / Grain Aphid

MATCH SCORE

~95%

Visual Match

Clinical Overview:

The image clearly shows a large colony of small, soft-bodied green aphids feeding along the wheat leaf. **This is an insect pest infestation, not a fungal disease.**

Aphids suck plant sap and can cause yellowing, curling, reduced growth, and yield loss when populations become high. Some aphid species can also transmit plant viruses (e.g. Barley Yellow Dwarf Virus - BYDV).

What to Check on Your Crop (Diagnostic Indicators)

Observed Sign	Diagnostic Interpretation
Many small green, soft-bodied insects clustered together	Aphid infestation is highly likely
Aphids concentrated along the leaf	Typical feeding behavior
Some winged adults among wingless aphids	Indicates the colony can spread to other plants
Leaf yellowing or weakening	Can occur with heavy sap feeding
Ants around aphid colonies	May indicate aphids (ants feed on secreted honeydew)

Treatment Protocol

Phase 1: Immediate Steps

- **1. Inspect the entire field:** Check whether aphids are confined to isolated patches, field edges, or are widely distributed across the wheat crop.
- **2. Check the undersides of leaves and stems:** Aphids often cluster along leaf veins, sheaths, and near developing heads.
- **3. Look for natural predators / beneficial insects:** Check for ladybugs (ladybird beetles), lacewing larvae, hoverfly larvae, or parasitized mummies before spraying insecticides.
- **4. Avoid unnecessary broad-spectrum spraying:** Overusing broad-spectrum insecticides can eliminate beneficial natural predators and exacerbate aphid outbreaks later.
- **5. Assess the growth stage of the wheat:** Aphid control is particularly critical from boot stage through grain-filling when yield impact is greatest.

Phase 2: Management & Chemical / Biological Interventions

Economic Threshold Guidelines: If aphid populations exceed local threshold guidelines (e.g. 5-10+ aphids per tiller/head depending on growth stage and local advisories):

- Consider an approved targeted aphid insecticide recommended by local agricultural extension services.
- Choose selective insecticides whenever possible to help conserve natural enemies.
- Always follow the label instructions regarding dosage, safety equipment, application timing, and pre-harvest interval (PHI).
- Rotate insecticide classes / modes of action (IRAC groups) to prevent aphids from developing insecticide resistance.
- Encourage beneficial insect populations by planting flowering border strips or avoiding preventive prophylactic calendar sprays.

Future Agronomy Insights & Key Prevention Rules

Aphid numbers can multiply very quickly under moderate temperatures. In wheat, early feeding on flag leaves and developing heads can reduce grain number and grain weight. In addition to direct feeding damage, some aphids can transmit Barley Yellow Dwarf Virus (BYDV), making regular monitoring of field borders essential.

- Regular field scouting → **Early detection**
 - Protect flag leaf and wheat heads → **Preserve grain yield**
 - Conserve beneficial insects → **Natural pest suppression**
- Spray only above economic thresholds → **Prevent resistance**
 - Rotate insecticide modes of action → **Avoid chemical failure**
 - Monitor field borders first → **Catch migrations early**

Important Field Distinction: Aphids vs. Diseases

Issue / Pest / Disease	Visible Presentation in Field
Wheat Aphids	Small, live, moving green insects clustering along leaves and stems; sticky honeydew
Leaf Rust	Orange/brown powdery fungal pustules scattered randomly across the leaf
Loose Smut	Entire wheat head replaced by a dark black/brown powdery spore mass
Powdery Mildew	White, fluffy/powdery fungal patches on the upper surface of leaves

***Diagnostic Notice:** Exact aphid species (such as English grain aphid, bird cherry-oat aphid, or greenbug) is best confirmed through closer microscopic examination of body shape and cornicles. Recommendations should be aligned with local agricultural extension guidelines.*